CDI COLLEGE

ASSIGNMENT 2

Technical Analysis Using R

Preliminary Stage

Student Name	Sara Mohamed Mohamed Flefla
Course	BDA400 – Data Science Tools and Techniques
Assignment	CA_BDA400_A02
Portfolio Symbols	AMZN, GOOGL, TSLA
Submission Date	November 12, 2026

GitHub Repository
Sara's exact repository URL must be added after confirmation.

1. Introduction and Objectives

This report documents the preliminary stage of a technical-analysis project using R: environment setup, packages, portfolio configuration, data importing, statistics, data display, and visualizations. Objectives include creating the TechnicalAnalysis repository; installing R/RStudio and packages; creating portfolio.txt; importing stock data; calculating the required statistics; and documenting the results.

2. GitHub Account and Repository

The assignment requires a TechnicalAnalysis repository, an Assignment2 subfolder, and a repository link. The RStudio working folder is Flefla_Sara_CA_BDA400_A02_package. Sara's exact GitHub URL has not been confirmed and is not guessed. Add genuine screenshots of the account, repository, and Assignment2 folder before submission.

3. Installing and Configuring R and RStudio

RStudio was opened successfully on Windows. The Console displayed R version 4.6.2. The working directory was C:/Users/ADMIN/Downloads/Flefla_Sara_CA_BDA400_A02_package. Current evidence confirms an active RStudio session, but not the original download wizard or every installation step. Add those genuine screenshots if required.

4. Installing Required R Packages

The project uses quantmod for market-data retrieval, TTR for technical-analysis calculations, xts for time-indexed data, and ggplot2 for charts. The script checks for and loads these packages. Add a genuine Console screenshot showing successful installation/loading.

5. Project Configuration and Portfolio

The portfolio.txt file contains one ticker symbol per line: **AMZN**, **GOOGL**, and **TSLA**. The script reads these symbols and stores the successfully imported time-series objects in portfolio_data.

6. Function to Import Stock Data

The load_stock_data() function reads portfolio.txt, removes blank entries, and retrieves data through quantmod. Each imported stock object is stored separately in portfolio_data. The Console confirmed that names(portfolio_data) returned AMZN, GOOGL, and TSLA.

7. Function to Compute Basic Statistics

The analysis calculates the latest 20-day simple moving average, mean, mode, median, and standard deviation using each stock's closing prices. The values visible in the captured RStudio summary output are:

Symbol	20-day MA	Mean	Mode	Median	Std. Dev.
AMZN	255.8675	241.9198	222.69	244.85	19.90911
GOOGL	308.1835	333.8298	318.58	336.02	28.88584
TSLA	381.8350	395.0312	372.11	395.23	38.02328

These values reproduce the recorded run and may change when the script is run again because market prices and available data can change.

8. Displaying Data and Visualizations

The script generates closing-price line charts for all three symbols and saves them in Stock_Charts. The screenshot confirms the three PNG files and displays the TSLA Closing Price chart.

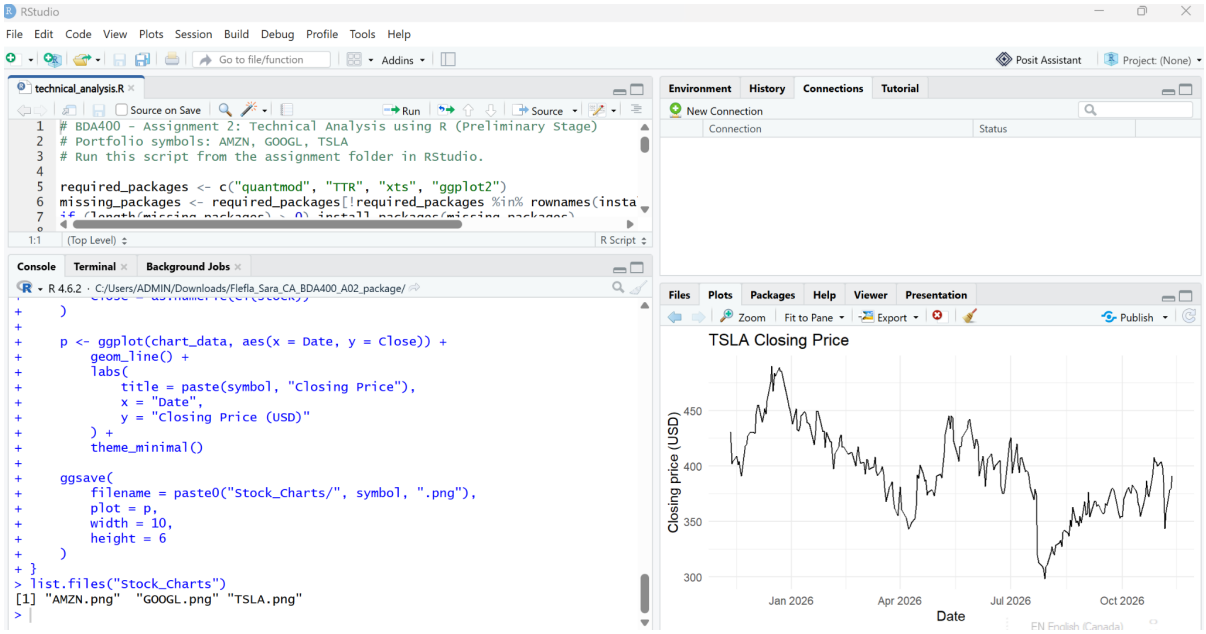


Figure 1. RStudio confirms AMZN.png, GOOGL.png, and TSLA.png; TSLA closing-price chart is displayed.

The script also exports the combined summary statistics to stock_statistics.csv. Confirm that the CSV file is present in the project folder and uploaded to GitHub.

9. How to Run the Project

1. Open the project folder in RStudio. 2. Confirm portfolio.txt and technical_analysis.R are in the working directory. 3. Run the script from top to bottom using Source. 4. Allow missing packages to install. 5. Review the Console for data and statistics. 6. Confirm the three PNG files in Stock_Charts. 7. Confirm stock_statistics.csv exists. 8. Upload all deliverables to Sara's TechnicalAnalysis/Assignment2 folder.

10. Project Deliverables

portfolio.txt; technical_analysis.R; stock_statistics.csv; Stock_Charts/AMZN.png; Stock_Charts/GOOGL.png; Stock_Charts/TSLA.png; README.txt or README.md; and this report in DOCX/PDF format.

11. Conclusion

The R workflow for Sara's portfolio has been executed: AMZN, GOOGL, and TSLA were loaded into portfolio_data, summary statistics were calculated, and three closing-price chart files were generated. The repository URL and remaining setup screenshots must be added before final submission.

12. Final Submission Checklist

Repository URL: Add Sara's confirmed TechnicalAnalysis repository link here.

Requirement	Evidence / Status
GitHub repository and URL	Needs confirmation / screenshot
Assignment2 uploaded	Needs GitHub screenshot
R and RStudio	RStudio running; installation evidence still needed
Required packages	Script uses/checks packages; add Console evidence
portfolio.txt	AMZN, GOOGL, TSLA
technical_analysis.R	Present in working folder
stock_statistics.csv	Confirm file exists
Three chart images	Confirmed: AMZN.png, GOOGL.png, TSLA.png

Requirement	Evidence / Status
MS Word report with screenshots	Prepared; add remaining genuine evidence

13. Additional Evidence – Project Folder and Saved Output

The screenshot below shows the local project folder Flefla_Sara_CA_BDA400_A02_package. It confirms that the folder contains stock_statistics.csv, the assignment PDF report, the portfolio text file, README, technical_analysis R script, and Stock_Charts folder.

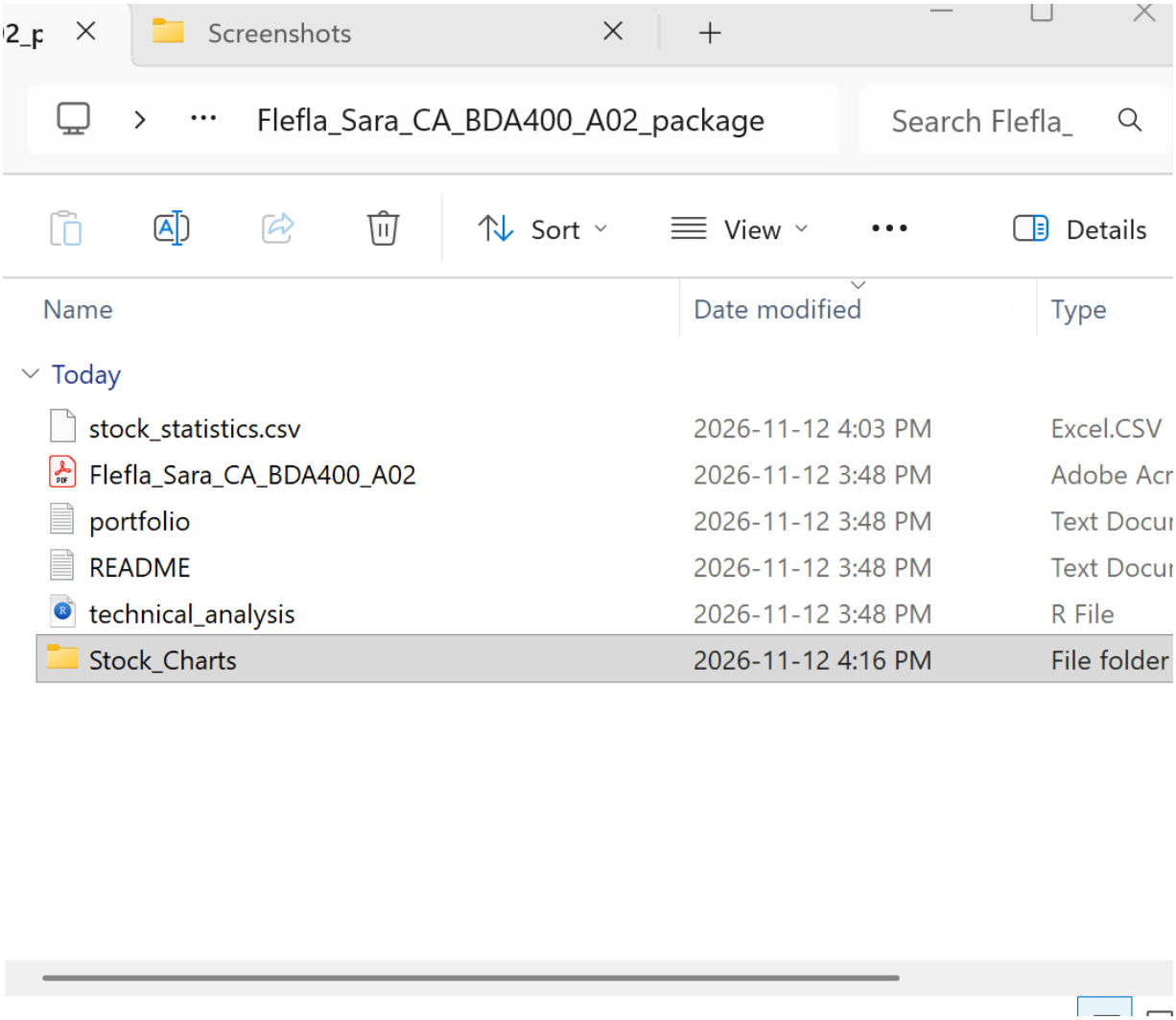


Figure 2. Local Assignment 2 project folder showing stock_statistics.csv and the project files.

Evidence update: stock_statistics.csv is confirmed as present in the local project folder. This screenshot does not prove that the files have been uploaded to GitHub. Sara's GitHub repository link and upload evidence still need to be added or verified.

14. RStudio Installation Evidence

The following screenshots document the RStudio installation process on the computer used for this assignment. They show the setup welcome screen, the selected installation folder, the installation in progress, and the successful completion message.

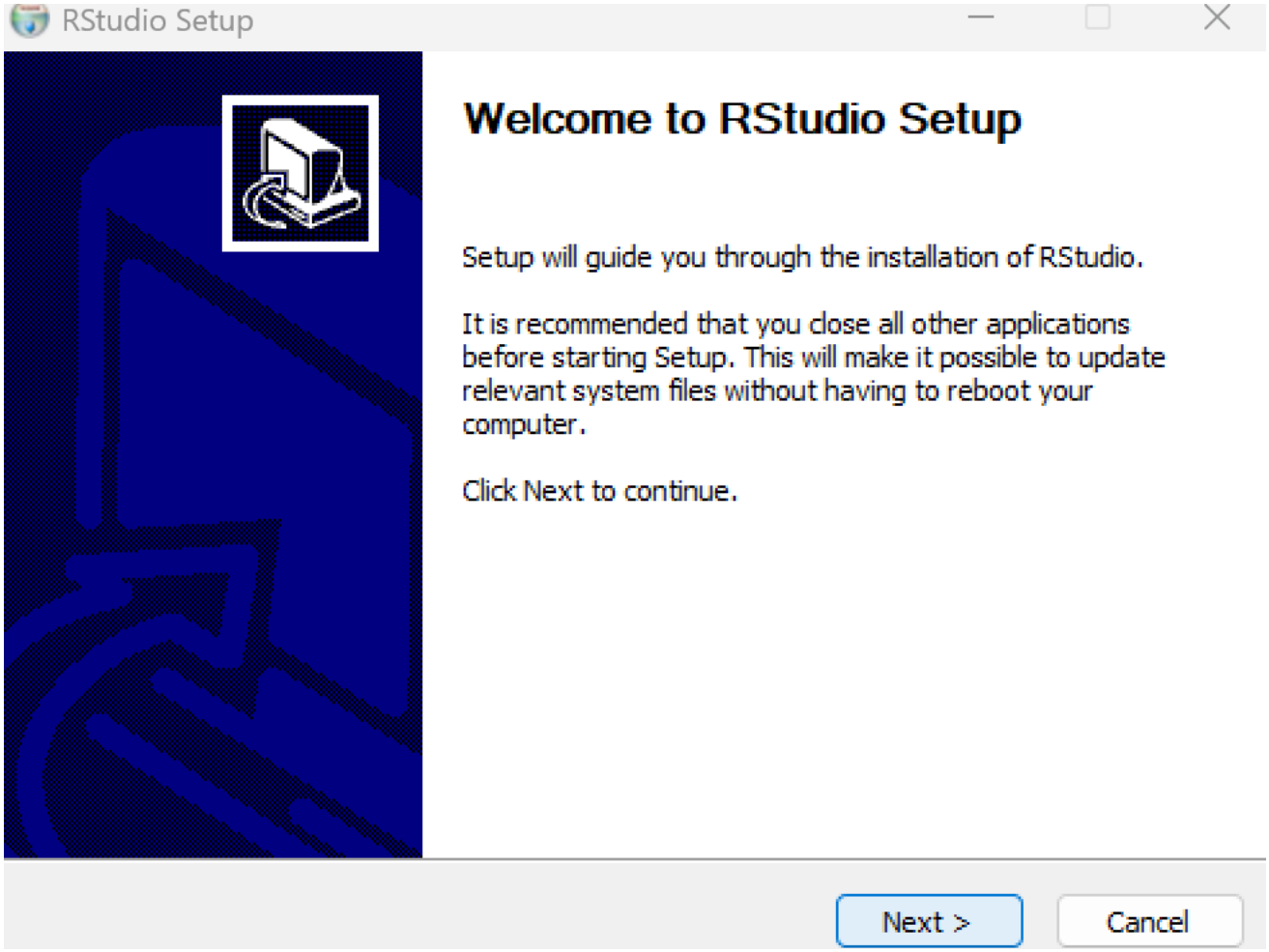


Figure 3. RStudio Setup welcome screen.

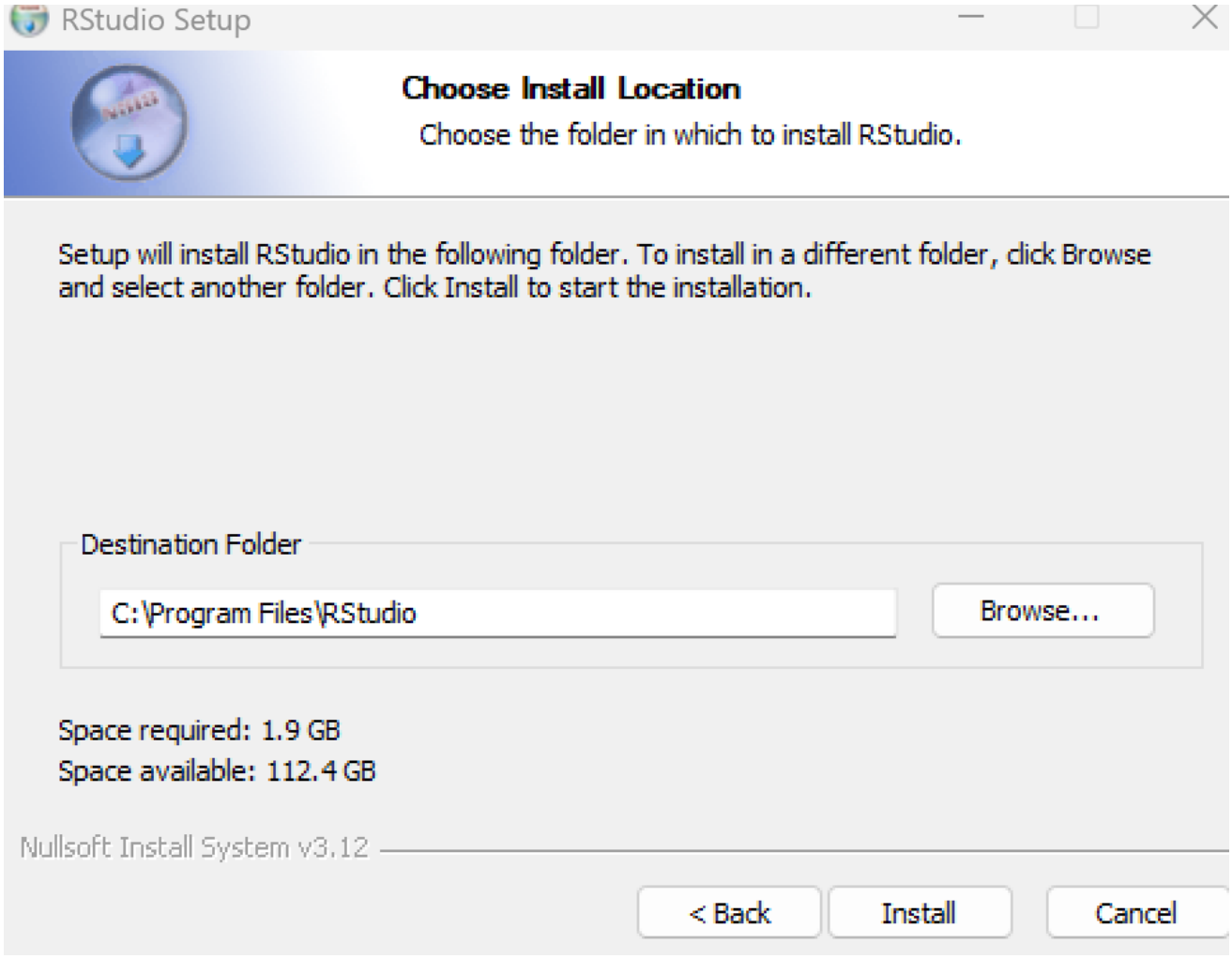


Figure 4. RStudio installation location selected.

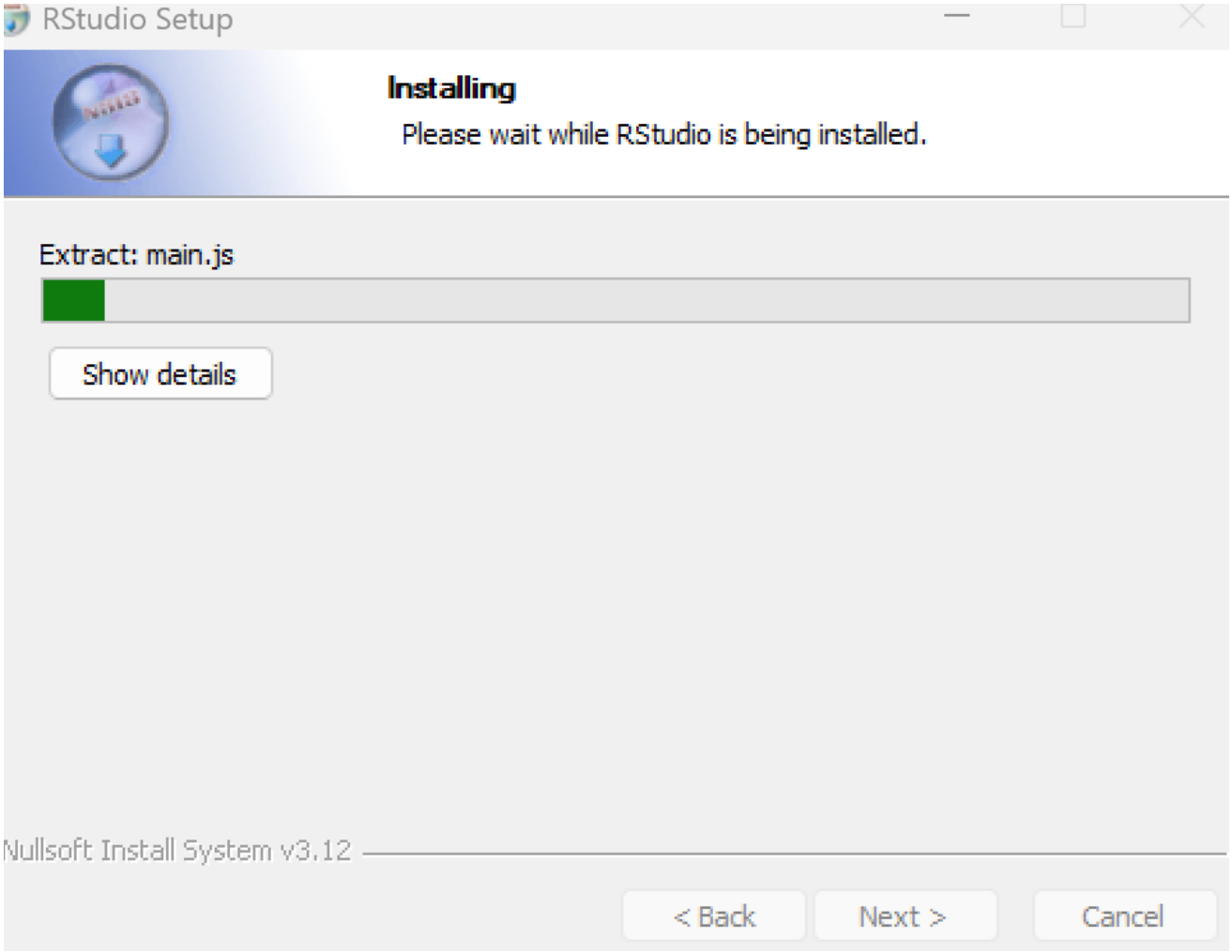


Figure 5. RStudio installation in progress.

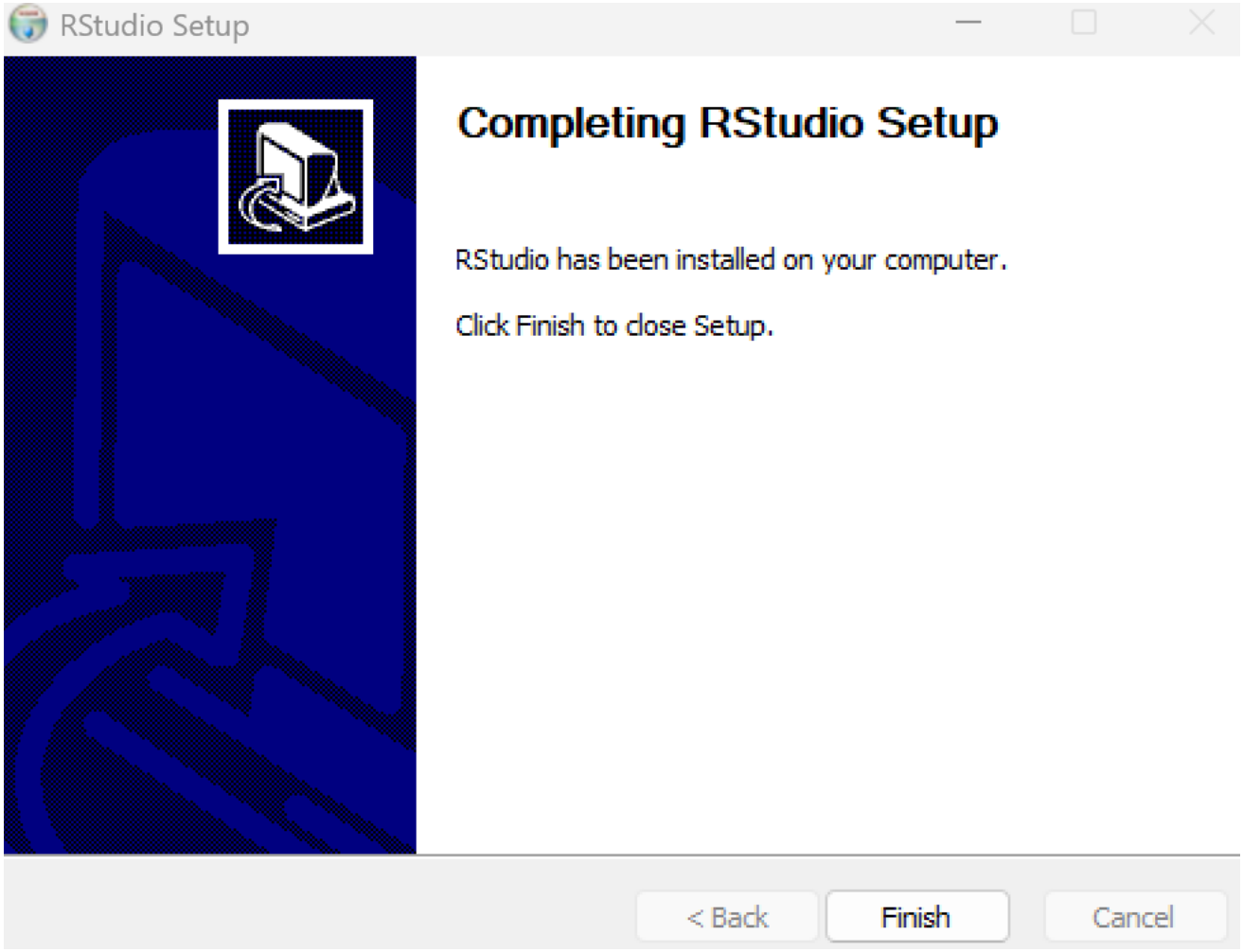


Figure 6. RStudio installation completed successfully.

Evidence note: The final setup screen states that RStudio has been installed on the computer. These screenshots establish installation only; they do not independently verify GitHub upload or assignment submission.